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Experiment – 13 : Case Study on Low-Code Limitations

Aim

To study the limitations of low-code application development using Zoho Creator and evaluate how these limitations may affect the development, customization, integration, performance, and scalability of the Event Desk event management platform.

Theory

Low-code development platforms allow applications to be created using visual development tools, predefined components, forms, workflows, and limited amounts of code. Zoho Creator supports rapid application development through visual interfaces, Deluge scripting, reports, dashboards, and integration features.

Although low-code platforms simplify application development, they may introduce limitations when applications require advanced customization, complex processing, specialized integrations, or extensive control over infrastructure.

The following limitations are relevant to the Event Desk platform:

1. Limited UI Customization

Visual page builders provide predefined layouts and components. Highly specialized user interfaces or complex interactive elements may require custom code or external technologies.

2. Platform Dependency

Applications built using Zoho Creator depend on its development environment, supported features, scripting language, and deployment model. Moving the entire application to another platform may require redesigning some components.

3. Deluge Scripting Limitations

Deluge supports application logic and workflow automation, but developers may encounter limitations when implementing advanced algorithms, specialized libraries, or complex processing requirements.

4. Integration Constraints

External services can be integrated using connections and APIs. However, integration may require additional authentication configuration, permission management, request handling, and troubleshooting.

5. Performance and Scalability

Application performance may depend on record volume, report complexity, workflow execution, API requests, and platform limits. Large datasets or frequent automation may require optimization and careful evaluation.

6. Limited Infrastructure Control

Low-code platforms generally manage much of the underlying infrastructure. Developers may have less control over servers, runtime environments, deployment architecture, and low-level system configurations than in a fully custom application.

7. Platform-Specific Data and Logic

Forms, reports, workflows, and Deluge scripts may depend on Zoho Creator-specific components. Recreating these features in another environment may require translating business logic and restructuring data.

8. Debugging and Troubleshooting

Visual configurations can make development easier, but errors involving workflow triggers, field link names, permissions, or API connections may still require detailed investigation.

9. Security and Governance Constraints

Security depends on the controls supported by the platform and their correct configuration. Highly specialized security requirements may need additional integration, monitoring, or architectural decisions.

10. Vendor Lock-In and Migration Effort

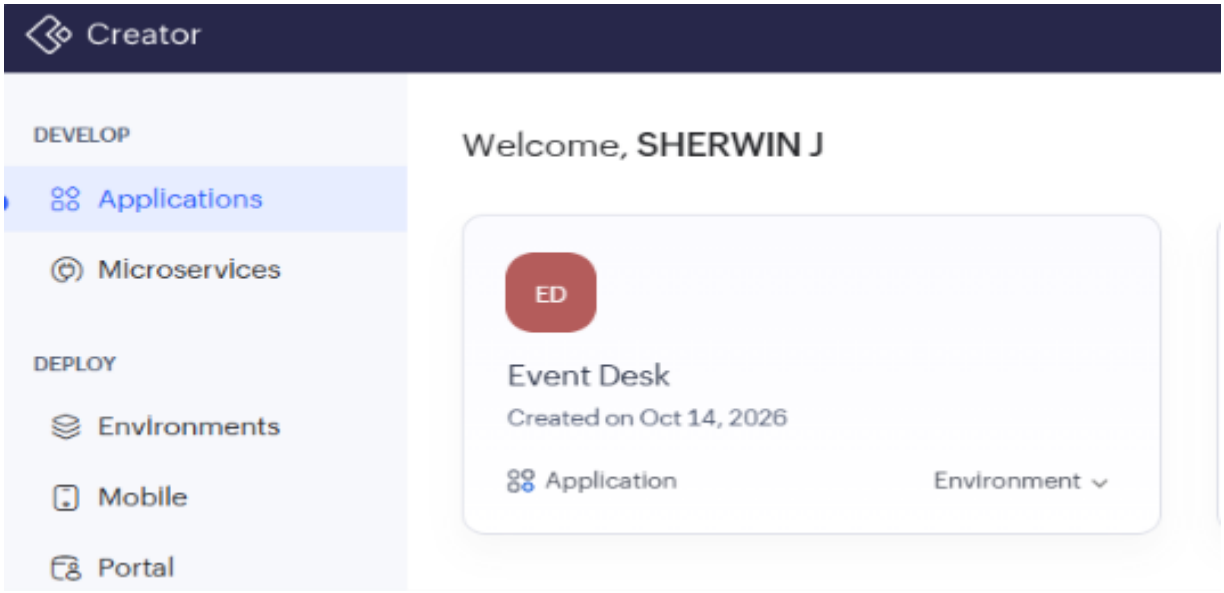
An application that relies heavily on platform-specific features may require substantial work to migrate. Exporting application definitions does not necessarily transfer every configuration, integration, credential, or runtime behavior automatically.

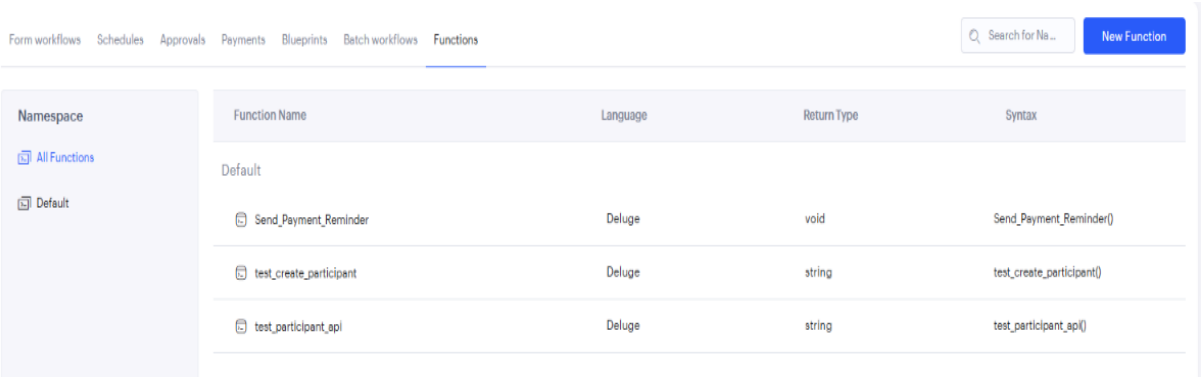
The Event Desk platform provides a practical case study because it includes registration forms, reports, dashboards, Deluge workflows, user management, and API integrations. These components demonstrate both the development advantages and the possible constraints of low-code technology.

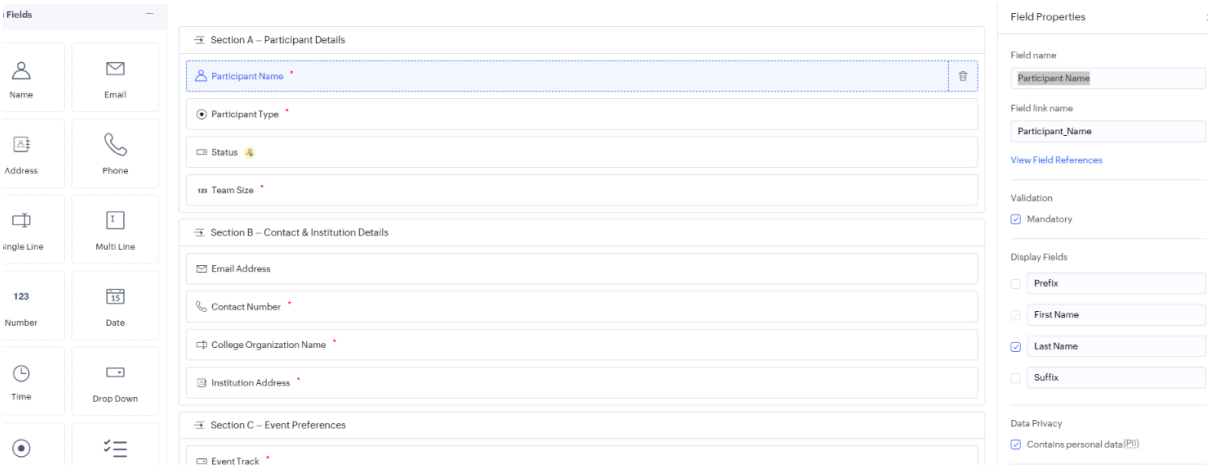
Procedure

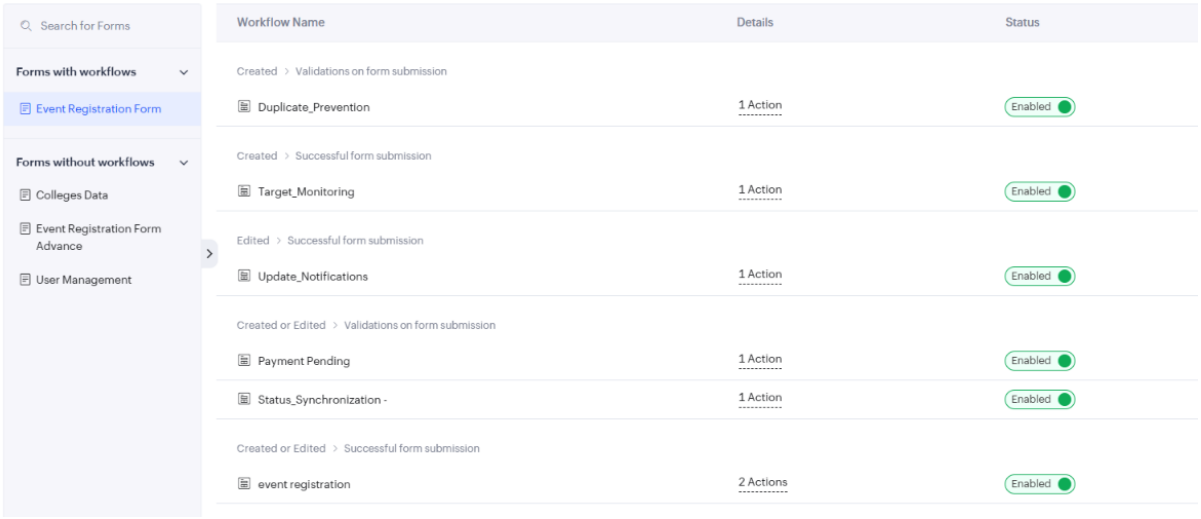
1. Open the Event Desk application in Zoho Creator.
2. Review the existing forms, including Event_Registration_Form.
3. Examine the Event_Registration_Report and identify the available reporting and customization options.
4. Review the Registration Status Board, Participation Dashboard, and AI Registration Analytics Dashboard.
5. Identify which interface elements can be configured through visual tools and which may require additional development.
6. Review the six workflows and examine their dependence on Deluge scripts, field names, conditions, and triggers.
7. Inspect the API connection and associated functions to understand the process of integrating external services.
8. Identify any platform-specific components that would need to be redesigned if the application were moved to another technology.
9. Review the application's data structure and consider the challenges of transferring participant records while preserving data integrity.
10. Examine available platform limits and configuration constraints relevant to records, workflows, API requests, and reports.
11. Consider how increased participant volume could affect report loading, dashboard visualization, and workflow execution.
12. Identify requirements that may be difficult to implement entirely through low-code features.
13. Compare the benefits of rapid development with the limitations involving customization, infrastructure control, and portability.
14. Document the limitations identified and determine whether they can be addressed through configuration, Deluge scripting, API integration, or external development.
15. Summarize the suitability of Zoho Creator for the current application and identify situations where additional technologies may be beneficial.

Screenshots









Result

The case study examined the benefits and limitations of using Zoho Creator to develop the Event Desk application. Low-code development simplifies form creation, reporting, dashboard design, and workflow automation. However, advanced customization, platform dependency, integration complexity, infrastructure control, and migration requirements may present challenges. The study shows that low-code platforms are suitable for many business applications, while complex or highly specialized requirements may benefit from additional pro-code development and external integrations.